

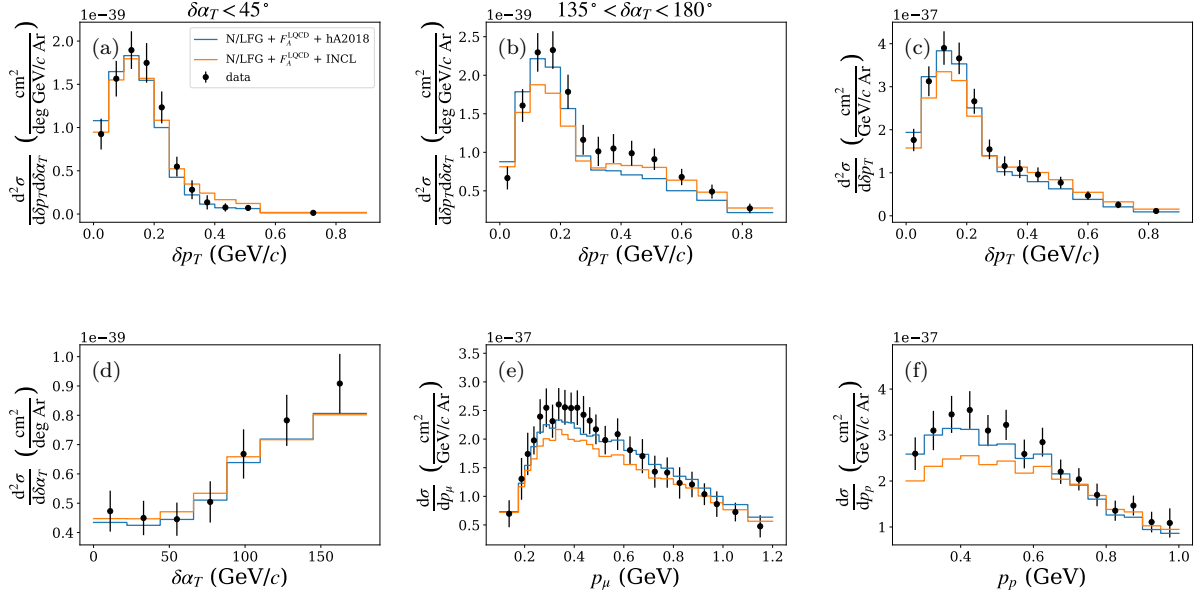


Figure 1: Effect of the final-state-interaction (FSI) model on the MicroBooNE CC1 μ 1p / CC1 μ Np argon cross sections with the N/LFG ground state and LQCD axial form factor held fixed: hA2018 (blue) versus INCL (orange), compared with MicroBooNE data (black points). Panels: (a) δp_T for $\delta\alpha_T < 45^\circ$ and (b) for $135^\circ < \delta\alpha_T < 180^\circ$ (two slices of the 2D δp_T - $\delta\alpha_T$ measurement); (c) δp_T ; (d) $\delta\alpha_T$; (e) p_μ ; (f) p_p . hA2018 agrees better with the data; switching to INCL raises χ^2/ndf in most observables (most strongly p_p ; see Table 1).

Table 1: χ^2/ndf (p -value) relative to MicroBooNE data for Fig. 1 (N/LFG ground state and LQCD axial form factor held fixed).

Configuration	δp_T vs. $\delta\alpha_T$	δp_T	$\delta\alpha_T$	p_μ	p_p
N/LFG + F_A^{LQCD} + hA2018	35.98/49 (0.92)	6.78/13 (0.91)	3.47/7 (0.84)	16.29/26 (0.93)	16.05/15 (0.38)
N/LFG + F_A^{LQCD} + INCL	56.01/49 (0.23)	9.09/13 (0.77)	5.75/7 (0.57)	17.66/26 (0.89)	26.94/15 (0.03)

Table 2: χ^2/ndf (p -value) relative to MicroBooNE data for Fig. 2 (spectral-function ground state and LQCD axial form factor held fixed).

Configuration	δp_T vs. $\delta\alpha_T$	δp_T	$\delta\alpha_T$	p_μ	p_p
SF + F_A^{LQCD} + hA2018	34.25/49 (0.95)	6.36/13 (0.93)	5.19/7 (0.64)	18.82/26 (0.84)	15.59/15 (0.41)
SF + F_A^{LQCD} + INCL	66.61/49 (0.05)	15.72/13 (0.26)	5.47/7 (0.60)	23.91/26 (0.58)	32.25/15 (0.01)

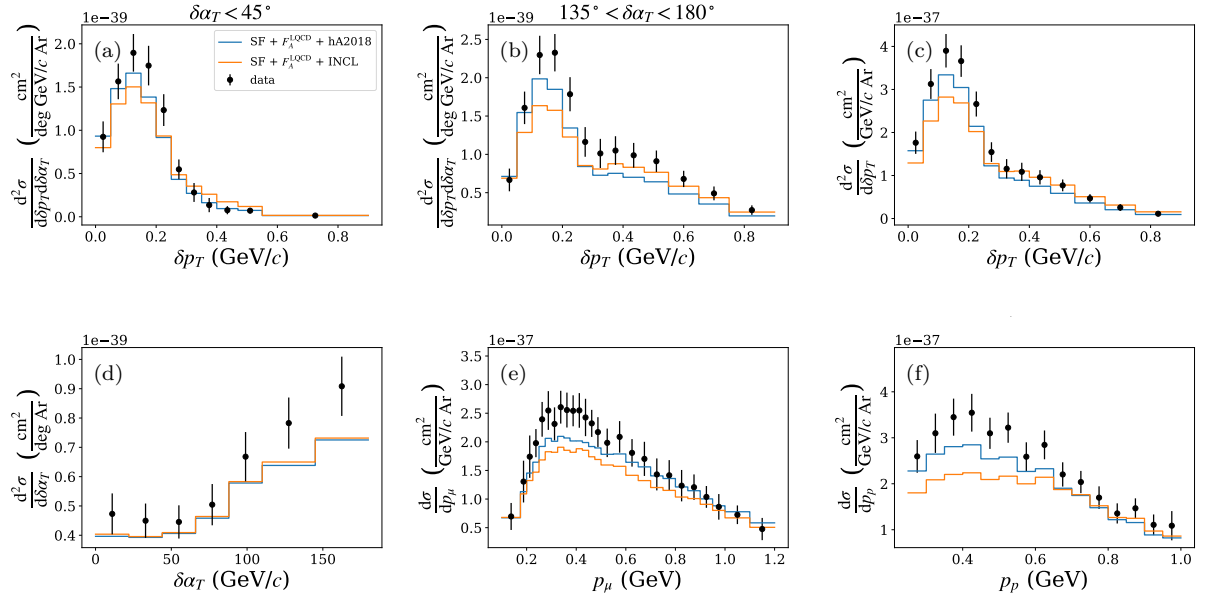


Figure 2: As in Fig. 1 but with the spectral-function ground state (LQCD axial form factor held fixed): hA2018 (blue) versus INCL (orange). Again hA2018 is favoured over INCL, which substantially worsens the 2D δp_T - $\delta\alpha_T$ and p_p agreement (see Table 2).